IT481 Capstone/Cornerstone Project Report on

Intelligent 3DCP: Layer Deformation Monitoring and Shrinkage Prediction Using ML

Submitted in partial fulfilment of the requirements for the degree of

by

BACHELOR of Technology in INFORMATION TECHNOLOGY

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DECLARATION

We hereby declare that the IT481 Capstone/Cornerstone Project Report entitled **"Intelligent 3DCP: Layer Deformation Monitoring and Shrinkage Prediction Using ML"**, which is being submitted to National Institute of Technology Karnataka, Surathkal, for the award of Degree of *Bachelor of Technology in Information technology, and Bachelor of Technology in Artificial Intelligence*, is a *bona fide* report of the work carried out by us. The material contained in this Report has not been submitted to any University or Institution for the award of any degree.

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CERTIFICATE

This is to certify that the Major Project Work entitled **"Intelligent 3DCP: Layer Deformation Monitoring and Shrinkage Prediction Using ML"** submitted by

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is the record of the work carried out by him/her/them, is accepted as the *B.Tech Capstone/Cornerstone project* work report submission in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Information Technology*(Sagar Hiremath)* and Bachelor of Technology in Artificial Intelligence*(Jeetu Kumar)* in the Department of Information Technology, NITK Surathkal.

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Intelligent 3DCP: Layer Deformation Monitoring and Shrinkage Prediction Using ML

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***Abstract*—3D Concrete Printing (3DCP) is an emerging additive manufacturing technology that enables automated, layer-by-layer construction of complex concrete structures, reducing labor, material waste, and construction time. Despite its potential, 3DCP faces critical challenges related to layer stability, process control, and long-term durability due to shrinkage and crack formation. This study focuses on three key aspects: (1) determining the maximum number of layers that can be printed before lower layers deform, (2) developing a real-time machine learning-based system to monitor local deformations and adjust printing parameters dynamically, and (3) predicting shrinkage and crack initiation using process, material, and environmental data. By integrating experimental observations with predictive modeling and adaptive control, the proposed framework aims to enhance the reliability, structural performance, and efficiency of 3DCP, paving the way for its wider adoption in modern construction practices.**

***Index Terms*—3D Concrete Printing, Additive Manufacturing, Layer Stability, Machine Learning, Shrinkage Prediction, Crack Formation**

I. Introduction

Concrete is the most widely used construction material in the world, forming the backbone of modern civilization. From buildings and bridges to roads and dams, it is an indispensable component of urban infrastructure and global economic development. Despite its ubiquity and versatility, **conventional concrete construction methods** have remained largely unchanged for decades. These traditional processes are often time-consuming, labor-intensive, and susceptible to human error. Moreover, they rely heavily on formwork, which not only increases material consumption and costs but also restricts the architectural freedom of designers and engineers.

In recent years, the construction industry has been undergoing a digital transformation, driven by the need for greater efficiency, sustainability, and design innovation. Among the most promising developments is **3D Concrete Printing (3DCP)**, an advanced additive manufacturing process that fabricates concrete structures layer by layer through the controlled extrusion of a specially formulated cementitious material. By eliminating the need for formwork and manual placement, 3DCP significantly reduces labor costs, minimizes material waste, and accelerates construction timelines. Furthermore, it enables the creation of complex geometries, organic shapes, and customized architectural forms that are often impractical or impossible to achieve through conventional casting methods.

Research and industrial applications have demonstrated that 3DCP can potentially **reduce overall construction costs by 50%–60%**, while simultaneously enhancing sustainability by optimizing material use and reducing the carbon footprint of construction projects. However, realizing these benefits on a large scale requires a thorough understanding of the **interplay between material science, mechanical design, and process control**. The printed structure's quality depends not only on the concrete mix design but also on the precise calibration of printer parameters such as nozzle diameter, extrusion pressure, print speed, and layer height.

During a recent visit to the **3DCP Laboratory at the National Institute of Technology Karnataka (NITK), Surathkal**, we observed the complete working mechanism of a large-scale 3D concrete printer. The visit provided valuable insights into the key **factors influencing the printing process**, including the rheological behavior of the printable mix, the bonding strength between consecutive layers, and the influence of environmental conditions such as temperature and humidity. It was evident that achieving a successful print is not solely dependent on advanced hardware but equally on the **optimization of material properties and real-time control of process parameters**.

One of the **major challenges** in 3DCP lies in developing an ideal **mix design** that meets multiple performance requirements simultaneously. The fresh concrete must possess adequate **pumpability** and **extrudability** to ensure smooth and continuous flow through the nozzle without clogging. At the same time, it must maintain sufficient **buildability** and **yield stress** to support subsequent layers without excessive defor-

mation or collapse. Achieving this delicate balance requires fine-tuning of mix proportions, admixtures, and additives that influence the setting time, thixotropy, and interlayer adhesion.

Furthermore, **printer parameters** such as extrusion pressure, nozzle travel speed, and interlayer time intervals play a critical role in determining the dimensional accuracy and mechanical integrity of the printed structure. Small deviations in these parameters can lead to layer misalignment, poor bonding, or structural instability. To overcome such issues, modern 3DCP systems are increasingly integrating **machine learning and real-time monitoring techniques** to adaptively control the printing process and predict material behavior under varying conditions.

In this context, the present study focuses on three critical aspects of 3D Concrete Printing: (1) ensuring layer stability during the printing process to prevent collapse or misalignment, (2) developing a real-time adaptive control system using machine learning for process optimization, and (3) predicting shrinkage and crack formation in printed elements to improve long-term durability. Addressing these challenges will contribute toward the advancement of reliable, automated, and sustainable 3D concrete construction technologies capable of reshaping the future of the construction industry.



Fig. 1. Example of a 3D Concrete Printing setup.

II. Background

The conventional construction industry has long faced persistent challenges, including low labor efficiency, high accident rates, a diminishing skilled workforce, and rising costs associated with time-consuming processes.[1] A significant portion of these costs and delays is attributable to the creation and removal of formwork, a necessary but wasteful and labor-intensive step in traditional concrete casting.[2] In response to these issues, 3D Concrete Printing (3DCP) has emerged as a paradigm-shifting technology, leveraging digital fabrication to automate construction processes.[3] As a form of additive manufacturing, 3DCP builds structures by depositing a specially formulated cementitious material layer-by-layer directly from a digital model. This method's primary advantage is the elimination of formwork, which offers the potential for accelerated construction timelines, reduced labor costs, minimized material waste, and unprecedented geometric freedom in architectural design.[2, 4]

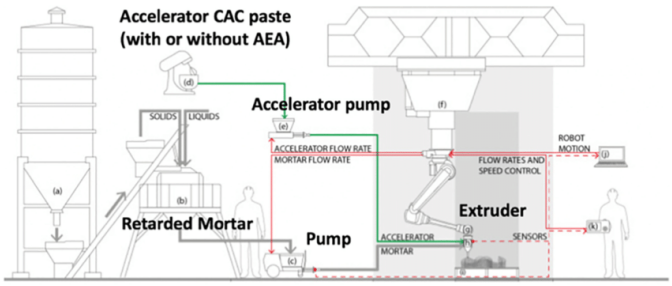


Fig. 2. 3D Concrete Printing Workflow

The foundational work in this field was pioneered through distinct, yet complementary, research streams that defined the technology's primary development paths. The seminal work by Behrokh Khoshnevis at the University of Southern California introduced *Contour Crafting*, a large-scale, on-site extrusion-based method designed for the rapid construction of entire buildings.[1] This approach laid the theoretical and technical groundwork for automated construction. A key innovation of Contour Crafting was the integration of robotic trowels adjacent to the extrusion nozzle, which smooth the surfaces of the deposited layers to create a high-quality, paint-ready finish directly within the printing process, while also allowing for the embedding of conduits for electrical and plumbing services [4].

Concurrently, a research team at Loughborough University, led by Richard Buswell, focused on an alternative philosophy: the off-site, factory-based prefabrication of highly complex and customized building components.[5] This method, termed *Concrete Printing*, emphasized precision and the integration of advanced manufacturing principles into construction. By using gantry systems and robotic arms in a controlled factory environment, this approach paved the way for sophisticated, engineered parts with intricate geometries that could be transported and assembled on-site [5]. Together, these pioneering efforts—one focused on holistic, on-site construction and the other on high-precision, off-site component manufacturing—established the primary technological pathways and demonstrated the transformative potential of 3DCP, catalyzing a global expansion of research and commercial application in the years that followed.

III. Problem Statement

The key problems addressed in this study are summarized as follows:

1) **Layer Stability:** Determining the maximum number of layers that can be printed before the lower layers begin to deform under the weight of subsequent layers.
2) **Adaptive Printing Control using Machine Learning:** Developing machine learning models to monitor local

deformations during printing and dynamically adjust printing speed or extrusion rate to maintain structural stability.

3) **Shrinkage and Crack Prediction:** Predicting shrinkage behavior and the onset of crack formation in printed structures to ensure long-term durability and interlayer bonding integrity.

IV. Dataset

The dataset for this study will primarily be obtained from the **Civil Engineering Department, NITK (3DCP Laboratory)**, which maintains experimental records of 3D concrete printing (3DCP) processes, material mix designs, and mechanical test data. To enhance data diversity and model generalization, supplementary data will also be incorporated from **open-source repositories and published studies** [1]–[3].

The dataset will comprise both *input features* (independent variables) and *output features* (dependent variables) relevant to layer stability, adaptive printing control, and shrinkage prediction.

A. Input Features

The input features encompass material composition, printing process parameters, and rheological properties that influence the mechanical and structural performance of printed layers.

- **Material Composition:**
  - Cement content (kg/m^3) primary binder influencing hydration rate and compressive strength [1].
  - Water content (kg/m^3) and water–binder ratio (w/b) control workability and layer adhesion [2].
  - Silica fume, fly ash, and slag (kg/m^3) supplementary cementitious materials that enhance microstructure and reduce shrinkage [3].
  - Fine and coarse aggregates (kg/m^3) and particle size (mm) influence rheology, buildability, and interlayer support.
  - Fiber content (kg/m^3), diameter (mm), and tensile strength (MPa) enhance crack resistance and flexural stability [1].
  - Viscosity-modifying agent and superplasticizer (kg/m^3) control extrudability, shape retention, and layer cohesion.
- **Process Parameters:**
  - Print speed (mm/s) affects extrusion uniformity and deformation rate under load [1].
  - Nozzle area (mm^2) and layer height (mm) determine flow rate and bonding surface area.
  - Interlayer time interval (s) influences adhesion strength and shrinkage cracks at layer interfaces.
  - Extrusion pressure (Pa) controls filament consistency and layer thickness stability [2].
  - Ambient temperature and relative humidity affect setting time and drying shrinkage [3].
- **Rheological and Mechanical Properties:**
  - Yield stress (Pa) and plastic viscosity (Pa·s) govern extrudability and build-up rate of printed layers.
  - Buildability index and deformation rate indicators of print stability and self-weight resistance.

B. Output Features

The dependent variables, representing the measurable responses of the printed concrete, will be used to evaluate model performance and predictive accuracy.

- **Compressive strength (MPa):** Indicates the load-bearing capacity and overall stiffness of printed layers.
- **Flexural strength (MPa):** Represents resistance to bending and interlayer failure [1].
- **Tensile strength (MPa):** Measures the interlayer bonding quality and crack resistance [2].
- **Layer deformation (mm):** Quantifies vertical settlement and layer displacement, used for real-time stability monitoring.
- **Shrinkage strain (%):** Captures both plastic and drying shrinkage responsible for dimensional instability [3].
- **Crack width and propagation rate (mm/s):** Measure structural integrity and durability of printed elements.

The integrated dataset will serve as the foundation for training and validating machine learning models capable of predicting deformation, shrinkage, and crack initiation in 3D-printed concrete, thereby supporting adaptive control mechanisms during the printing process.

V. Objectives

The primary objectives of this work are as follows:

1) **Maximum Layer Stability:** Determine the maximum number of layers that can be printed before lower-layer deformation exceeds an acceptable limit.
2) **Real-time ML-based Control:** Develop a real-time machine learning-based control system to detect local deformations and adjust print speed to minimize further distortions.
3) **Shrinkage and Crack Prediction:** Predict shrinkage and crack formation in printed layers using process, material, and environmental data.

The **overall objective** is to create an integrated 3DCP monitoring and control system that reduces deformations and crack occurrences during the printing process.

VI. Scope and Deliverables

A. Scope

The scope of this project is centered on the development of machine learning models to enhance 3D Concrete Printing (3DCP) by predicting and mitigating key challenges. Specifically, the work focuses on:

- Predicting the maximum number of layers that can be printed before lower layers experience deformation.
- Developing a machine learning model to predict shrinkage and crack formation in printed structures based on process, material, and environmental data.

- Providing insights from model predictions to guide optimal printing parameters and mix design decisions.

B. Deliverables

The expected deliverables for this ML-oriented project include:

- Curated datasets comprising layer deformation measurements, material properties, and environmental conditions.
- A predictive machine learning model for layer deformation.
- A predictive machine learning model for shrinkage and crack initiation.
- Analysis and visualizations of model predictions to inform decision-making for 3DCP.
- A detailed report documenting methodology, model development, results, and recommendations.

VII. Literature Review

The application of Machine Learning (ML) to address the inherent variabilities and challenges in 3D Concrete Printing (3DCP) is a burgeoning field of research. Predictive modeling, in particular, has shown significant promise in optimizing mix designs and forecasting the mechanical performance of printed structures, thereby reducing the need for extensive and time-consuming experimental trials.[6]

Several studies have focused on predicting the final hardened properties of 3DCP. For instance, Ali et al. [1] developed various ML models, including Support Vector Machine (SVM) and XGBoost Regression, to predict the flexural and tensile strength of 3D printed concrete.[7] Their work utilized a comprehensive dataset of mix design parameters such as water-to-binder ratio, cement content, and fiber properties as inputs. The study successfully demonstrated that the SVM model could predict the mechanical properties with high accuracy, establishing a strong precedent for using ML to navigate the complex, multi-variable problem of mix design optimization.[7]

Building on this approach, Zhang et al. [2] also employed a suite of ML algorithms, including random forests and artificial neural networks, to predict the mechanical properties of 3D printed concrete, achieving high correlation coefficients.[8] Their research further underscores the capability of ML to model the complex non-linear relationships between material composition and structural performance. More recent work has even extended this approach to predict critical process-related properties, such as the interlayer bonding strength (IBS), using advanced ensemble techniques like stacking models to improve predictive accuracy over single-model approaches.[9]

While these studies provide a robust framework for predicting final mechanical strength based on initial mix proportions, a significant gap remains in the real-time monitoring and prediction of dynamic behaviors during the printing process itself. The critical challenges of early-age layer deformation under loading from subsequent layers, and the long-term effects of shrinkage, are highly dependent not only on the mix design but also on dynamic process parameters (e.g., print speed) and ambient environmental conditions.[10] The existing literature has not fully addressed the development of adaptive ML systems that can predict these transient phenomena in real-time to actively control the printing process. This study aims to bridge that gap by developing intelligent models focused specifically on predicting layer deformation and shrinkage, integrating process and environmental data to enable a dynamic, adaptive control system for 3DCP.

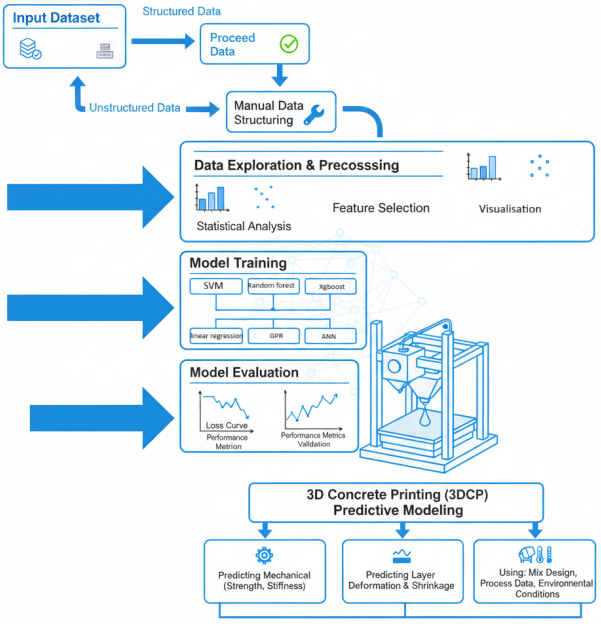


Fig. 3. ML Workflow.

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